SC4001 – Part B: Chest X-ray Classification

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Generative AI Declaration:

I used **Codex** to assist with the generation of selected code segments during the development of this assignment. The tool was used primarily to generate initial code templates and to obtain suggestions for implementing specific programming tasks. All AI-generated code was carefully reviewed, modified where necessary, integrated into the overall solution by me, and tested to ensure it met the assignment requirements. I verified the correctness of the generated code through manual inspection, compilation/execution testing, debugging, and comparison with the assignment specifications and relevant course materials. The final submitted work represents my own understanding and effort. I take full responsibility for the accuracy, functionality, and academic integrity of the submitted solution.

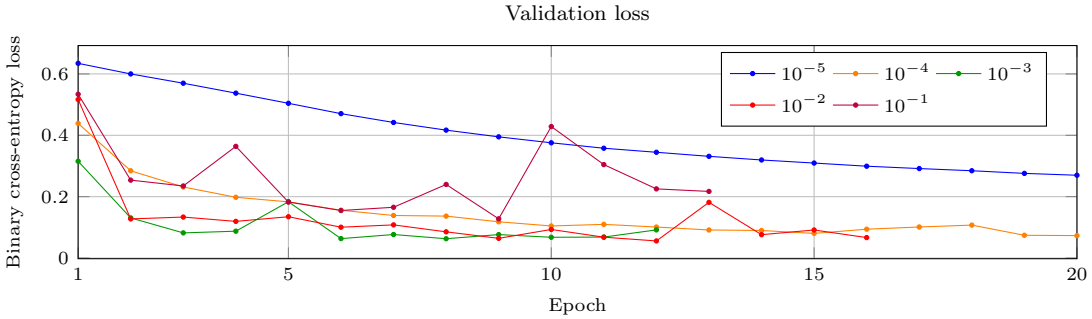
Generative AI Tool Used: Codex

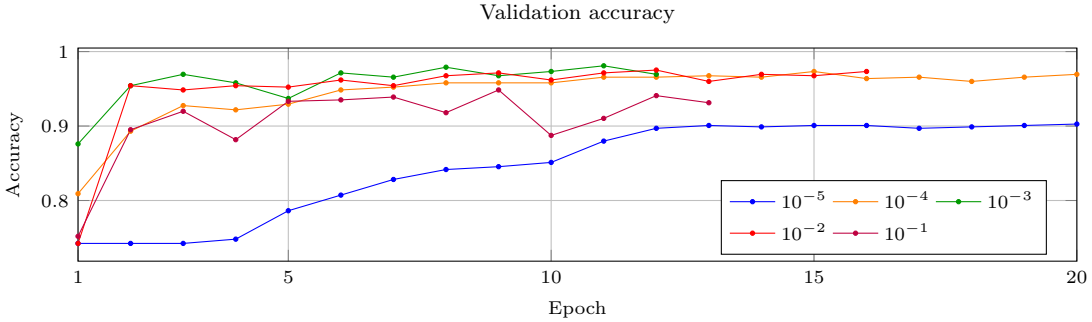
Purpose of Use: Code generation and programming assistance

Verification Method: Manual code review, testing, debugging, and validation against assignment requirements

B1

(a) Validation Loss and Accuracy Curve for Each Learning Rate





(b) Learning Rate Comparison Table

Pneumonia is positive (label 1); the prediction threshold is 0.5. TP, TN, FP, FN denote true positives, true negatives, false positives and false negatives; $N = TP + TN + FP + FN$.

$$\text{Accuracy} = \frac{TP + TN}{N}, \qquad \text{Error} = 1 - \text{Accuracy},$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}.$$

Validation loss is binary cross-entropy, with true label y_i and predicted pneumonia probability p_i:

$$\mathcal{L}_{\text{val}} = -\frac{1}{N} \sum_{i=1}^{N} \big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \big].$$

Each row reports the minimum-error checkpoint (ties: lower loss, then earlier epoch). Error, sensitivity and specificity are percentages.

Learning rate	Best epoch	Epochs run	Val. loss	Error (%)	Sens. (%)	Spec. (%)
10^{-5}	20	20	0.2698	9.73	96.14	73.33
10^{-4}	15	20	0.0803	2.67	97.69	96.30
10^{-3}	11	12	0.0683	1.91	97.94	98.52
10^{-2}	12	16	0.0557	2.48	98.20	95.56
10^{-1}	9	13	0.1278	5.15	96.92	88.89

(c) Optimal Learning Rate and Discussion

Optimal learning rate is 10^{-3} with the lowest validation error of 1.91%. The learning curves show that the model with learning rate 10^{-3} converges faster than other learning rates, with the best validation loss and accuracy in most of the epochs. Sensitivity and specificity are also needed to interpret model performance because they reflect the models' ability to detect pneumonia cases and correctly identify healthy subjects, respectively. And the learning rate 10^{-3} has high sensitivity and specificity, proving its effectiveness in accurate detection of both categories.

(d) Test Performance

The validation-selected checkpoint (10^{-3}, epoch 11) was evaluated once on 624 test images using training-set normalisation and a threshold of 0.5.

Accuracy (%)	Sensitivity (%)	Specificity (%)
87.02	99.49	66.24

The confusion counts were $TP = 388$, $TN = 155$, $FP = 79$ and $FN = 2$.

B2

(a) VGG-like Architecture and Training

(b) Validation Curves and Batch Size Comparison

(c) Optimal Batch Size and Discussion

(d) Test Performance

B3

(a) ResNet-like Architecture and Training

(b) Model Performance

(c) Architecture Comparison and Discussion

B4

(a) X-ray Selection and Grad-CAM Implementation

(b) Grad-CAM Overlays

(c) Comparison and Interpretation